

Backend Tasks v0.1

Backend Team Priority Order

1. Supabase project setup;
2. Database migrations;
3. Organisation and email-domain verification;
4. Authentication;
5. Profiles and interests;
6. Areas and availability;
7. Activities and join requests;
8. Official events and RSVPs;
9. Real-time map data;
10. Matching logic;
11. Connection requests;
12. Private and activity chat;
13. Block and report functionality;
14. RLS and security testing;
15. Staging deployment and documentation.

Progress tracker

	☰ Phases	🕒 Status	📅 End date	☰ Blockers
1	Phase 0 — Backend Foundation	☒ In progress		
2	Phase 1 — Authentication and Organisation Verification	☐ Not started	Select date	
3	Phase 2 — Profiles, Interests and Availability	☐ Not started	Select date	
4	Phase 3 — Real- Time Map, Activities and Events	☐ Not started	Select date	

5	Phase 4 — Matching and Connections	☐ Not started	Select date	
6	Phase 5 — Chat and Safety	☐ Not started	Select date	
7	Phase 6 — Testing, Deployment and Documentation	☐ Not started	Select date	

The backend must enforce these rules:

- Only verified students and professionals can use core features.
- Users must register with approved university or organisation emails.
- Exact personal GPS locations must never be exposed.
- Activities and events must expire automatically.
- Only authorised users can access profiles, chats and map data.
- Blocked users must not appear in suggestions, activities or map results.

Phase 0 — Backend Foundation

	✓ Task	👤 Assigned to	📅 Due date	🗑 Bucket
1	☐ Create Supabase development project	AF Ahmad El-Sheikh Fofol	Mon, Aug 31, 2026	Done
2	☐ Configure environment variables	AF Ahmad El-Sheikh Fofol	Sun, Aug 30, 2026	Done
3	☐ Set up Supabase migrations	AF Ahmad El-Sheikh Fofol	Sun, Aug 30, 2026	Done
4	☐ Define naming and branch conventions	AF Ahmad El-Sheikh Fofol	Sun, Aug 30, 2026	Done
5	☐ Create base organisation tables	AF Ahmad El-Sheikh Fofol	Sun, Aug 30, 2026	Done
6	☐ Create location tables	AF Ahmad El-Sheikh Fofol	Sun, Aug 30, 2026	Done
7	☐ Add seed data	AF Ahmad El-Sheikh Fofol	Sun, Aug 30, 2026	Done
8	☐ Define API response format	AF Ahmad El-Sheikh Fofol	Sun, Aug 30, 2026	Done
9	☐ Configure basic testing structure		Sun, Aug 30, 2026	To do

10	☐ Select map provider			To do
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Exit Criteria

- Supabase development environment is available.
- Migrations can be applied successfully.
- Initial organisations and areas are available.
- Environment variables are documented.
- Frontend team receives the first database and API documentation.

Phase 1 — Authentication and Organisation Verification

	✓ Task	👤 Assigned to	📅 Due date	📦 Bucket
1	☐ Configure Supabase Auth	 Ahmad El-Sheikh Folfol	Sat, Sep 5, 2026	In Progress
2	☐ Implement email verification	 Jay Nguyen	Thu, Sep 3, 2026	In Progress
3	☐ Validate approved email domains	 Jay Nguyen	Thu, Sep 3, 2026	In Progress
4	☐ Link users to organisations	 Jay Nguyen	Thu, Sep 3, 2026	In Progress
5	☐ Create profile on registration	 Jay Nguyen	Thu, Sep 3, 2026	In Progress
6	☐ Add account status fields	 Jay Nguyen	Thu, Sep 3, 2026	In Progress
7	☐ Restrict unverified users	 Jay Nguyen	Thu, Sep 3, 2026	In Progress
8	☐ Store terms and privacy consent	 Jay Nguyen	Thu, Sep 3, 2026	In Progress
9	☐ Test valid and invalid domains	 Jay Nguyen	Thu, Sep 3, 2026	In Progress
10	☐ Document authentication flow	 Jay Nguyen	Thu, Sep 3, 2026	In Progress

Required Rules

- `student@uts.edu.au` may register if the domain is approved.
- `person@gmail.com` must be rejected.
- An account cannot access matching, activities or chat before verification.
- The email address must not be visible to other users.

Exit Criteria

A student or professional can register, verify their email and be linked to the correct organisation.

Phase 2 — Profiles, Interests and Availability

Tasks

	✓ Task	👤 Assigned to	📅 Due date	📁 Bucket
1	☐ Create profile schema			
2	☐ Create interests schema			
3	☐ Create availability schema			
4	☐ Add profile CRUD operations			
5	☐ Add interest management			
6	☐ Add availability validation			
7	☐ Add approximate area selection			
8	☐ Add availability expiry logic			
9	☐ Implement profile RLS			
10	☐ Implement availability RLS			
11	☐ Add indexes			
12	☐ Write unit and integration tests			

Privacy Rules

- Store `area_id`, not exact user coordinates.
- Do not expose a user's email address.
- Do not expose a user's exact location.
- Users can only edit their own profile and availability.
- Expired availability must not appear on the map or in matching.

Exit Criteria

A verified user can create a profile, select interests, set availability and select an approximate area.

Phase 3 — Real-Time Map, Activities and Events

User-Created Activities

	✓ Task	👤 Assigned to	📅 Due date	📁 Bucket
1	☐ Create activities schema			
2	☐ Create participant schema			
3	☐ Implement activity creation			
4	☐ Implement activity editing			
5	☐ Implement activity cancellation			

6	☐ Implement join requests			
7	☐ Add participant limits			
8	☐ Add automatic activity expiry			
9	☐ Add activity RLS policies			
10	☐ Add activity report/block rules			

Official Events

	☒ Task	 Assigned to	 Due date	 Bucket
1	☐ Create official events schema		Select date	
2	☐ Create RSVP schema		Select date	
3	☐ Restrict event creation		Select date	
4	☐ Implement event creation		Select date	
5	☐ Implement event updates		Select date	
6	☐ Implement RSVP workflow		Select date	
7	☐ Add event approval status		Select date	
8	☐ Add automatic event expiry		Select date	
9	☐ Add event location data		Select date	
10	☐ Add event RLS policies		Select date	

Real-Time Map

	☒ Task	 Assigned to	 Due date	 Bucket
1	☐ Define map data format			
2	☐ Create active-map query			
3	☐ Add area-based filtering			
4	☐ Configure Supabase Realtime			
5	☐ Add event/activity channels			
6	☐ Prevent exact user-location exposure			
7	☐ Test map permissions			

Suggested Map Data Types

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- 1 presence
- 2 activity
- 3 official_event
- 4 public_venue

Activity Rules

- Only verified users can create activities.
- Activities must use a predefined area or public venue.
- Users cannot enter private addresses.
- Activities have a maximum participant limit.
- Activities automatically expire after the end time.
- Blocked users cannot view or join activities.

Exit Criteria

- Users can create small spontaneous activities.
- Approved organisations can publish official events.
- Activities and events appear on the real-time map.
- Expired items automatically disappear.
- Exact user locations are never exposed.

Phase 4 — Matching and Connections

	✓ Task	👤 Assigned to	📅 Due date	📁 Bucket
1	☐ Define matching rules			
2	☐ Create matching query/function			
3	☐ Add time-overlap calculation			
4	☐ Add area compatibility			
5	☐ Add shared-interest calculation			
6	☐ Exclude blocked users			
7	☐ Exclude existing connections			
8	☐ Create suggestions endpoint			
9	☐ Create connection requests			
10	☐ Add accept/decline/cancel actions			
11	☐ Add connection RLS			
12	☐ Test matching performance			

Matching Rules

A user can be recommended when:

- Both users are verified;
- They belong to an approved community;
- Their availability overlaps;
- Their approximate areas are compatible;
- They have at least one shared interest;
- There is no block relationship;
- They are not already connected.

Exit Criteria

Users can receive relevant suggestions and establish connections through the backend.

Phase 5 — Chat and Safety

	✓ Task	👤 Assigned to	📅 Due date	📁 Bucket
1	☐ Create messages schema			
2	☐ Implement private chat			
3	☐ Implement activity chat			
4	☐ Configure Realtime messages			
5	☐ Add message RLS			
6	☐ Implement block user			
7	☐ Implement report user			
8	☐ Add message rate limiting			
9	☐ Add request rate limiting			
10	☐ Test unauthorised access			
11	☐ Document moderation workflow			

Safety Rules

- Only connected users can use private chat.
- Only approved activity participants can use activity chat.
- Users cannot read another user's conversations.
- Blocking must remove map, activity and matching visibility.
- Reports must not be visible to the reported user.
- Exact GPS locations must never be stored for personal presence.

Exit Criteria

Connected users can chat privately, activity participants can coordinate, and users can block or report others.

Phase 6 — Testing, Deployment and Documentation

	✓ Task	👤 Assigned to	📅 Due date	🗑 Bucket
1	☐ Run full backend test suite			
2	☐ Audit all RLS policies			
3	☐ Test authentication edge cases			
4	☐ Test map privacy			
5	☐ Test activity and event expiry			
6	☐ Test matching logic			
7	☐ Test blocking and reporting			
8	☐ Configure staging environment			
9	☐ Configure production environment			
10	☐ Add monitoring and logs			
11	☐ Write backend README			
12	☐ Prepare frontend integration guide			

Exit Criteria

The backend supports the full MVP flow:

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1 Register

2 → Verify

3 → Create profile

4 → Set availability

5 → View map

6 → Create or join activity

7 → Match with users

8 → Connect

9 → Chat

10 → Block/report